LINFO2262: Gaussian Models

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Outline

1. Gaussian Model
   - Model Definition
   - Gaussian Model Estimation
   - Interpretation of the Covariance Matrix
   - Decision boundaries of a Gaussian classifier with 2 classes
2. Fisher Linear Discriminant
   - Model Definition
   - Relation to Gaussian Modeling

Outline

1. Gaussian Model
   - Model Definition
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   - Decision boundaries of a Gaussian classifier with 2 classes
2. Fisher Linear Discriminant

Motivation

MAP classifier is optimal but requires a good estimate of the likelihood $P(D|\mathscr{h})$

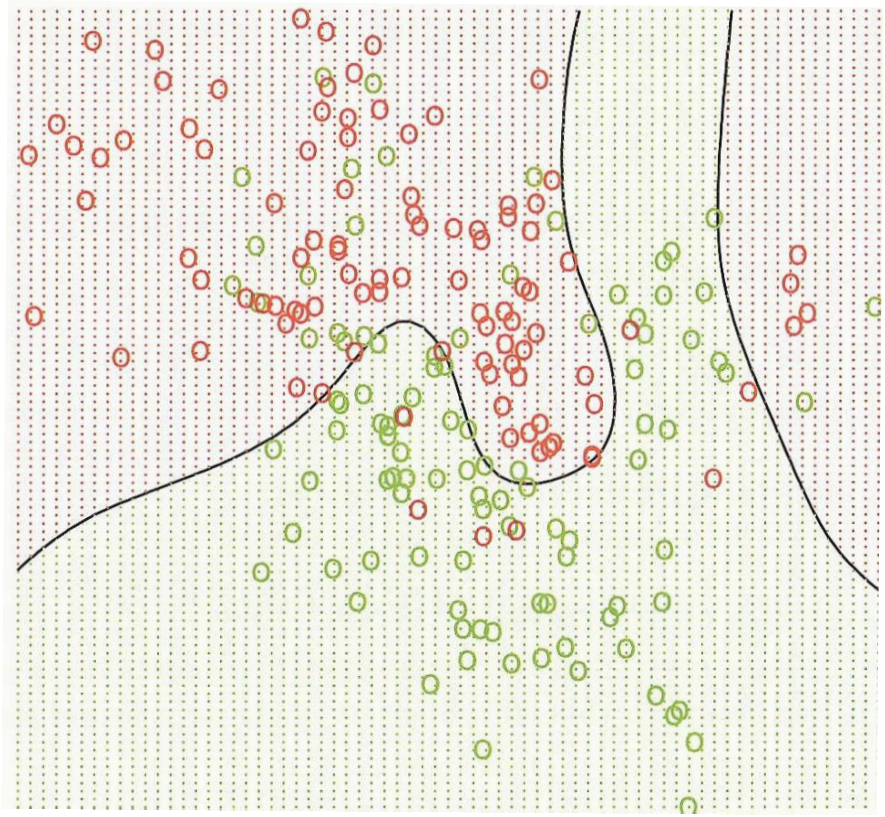


Illustration from *The Elements of Statistical Learning, (2nd edition)*, 2009, Springer.

How to tackle the estimation problem of $P(D|\mathcal{h})$?

Discriminative models

- Estimate globally decision boundaries (SVMs, Trees, Deep Nets. . .)
- Approximate locally decision boundaries with k-Nearest Neighbors

Generative models

- Assume conditional independence between the attributes (Naive Bayes)
- Assume the data follows some parametric (typically Gaussian) distributions

Gaussian Modeling

Assumptions:

- Target classification function $\mathbb{R}^d \rightarrow \{1, \ldots, C\}$, where each examples is described by a d-dimensional real vector $\boldsymbol{x}$
- All examples belonging to a given class are accurately modeled by a Gaussian distribution $N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$

Learning Procedure:

- Split the training data D into C subsets, where subset D_y contains all training examples labeled with a specific label y
- Estimate priors: $\hat{P}(y) = \frac{|D_y|}{|D|}$
- Estimate each class density f_y so as to maximize the data likelihood:
  - estimate mean vector: $\hat{\boldsymbol{\mu}}_y$ ($\Rightarrow d$ parameters per class)
  - estimate covariance matrix: $\hat{\boldsymbol{\Sigma}}_y$ ($\Rightarrow \mathcal{O}(d^2)$ parameters per class)

MAP classification of a new example $\boldsymbol{x}$:

$$\hat{y} = \underset{y \in \{1,\ldots,C\}}{\operatorname{argmax}} \; \hat{P}(y)\hat{f}_y(\boldsymbol{x})$$

Maximum likelihood estimation and classification

Learning

Maximum likelihood estimation (MLE):

$$\Rightarrow \text{compute } \hat{\Theta}_y = \text{argmax}_{\Theta}\, P(D_y|\mathscr{h}_y, \Theta_y)$$

- $D = \{D_1, \ldots, D_y, \ldots, D_C\}$, with D_y the training data for class y
- one model $\mathscr{h}_y$ represents each class y
- Θ_y is the set of parameters of $\mathscr{h}_y$

Repeat the estimation for each class $y \in \{1, \ldots, C\}$

Classification

MAP classification of a new $\boldsymbol{x}$ reduces to maximum likelihood classification when class priors are assumed to be equal:

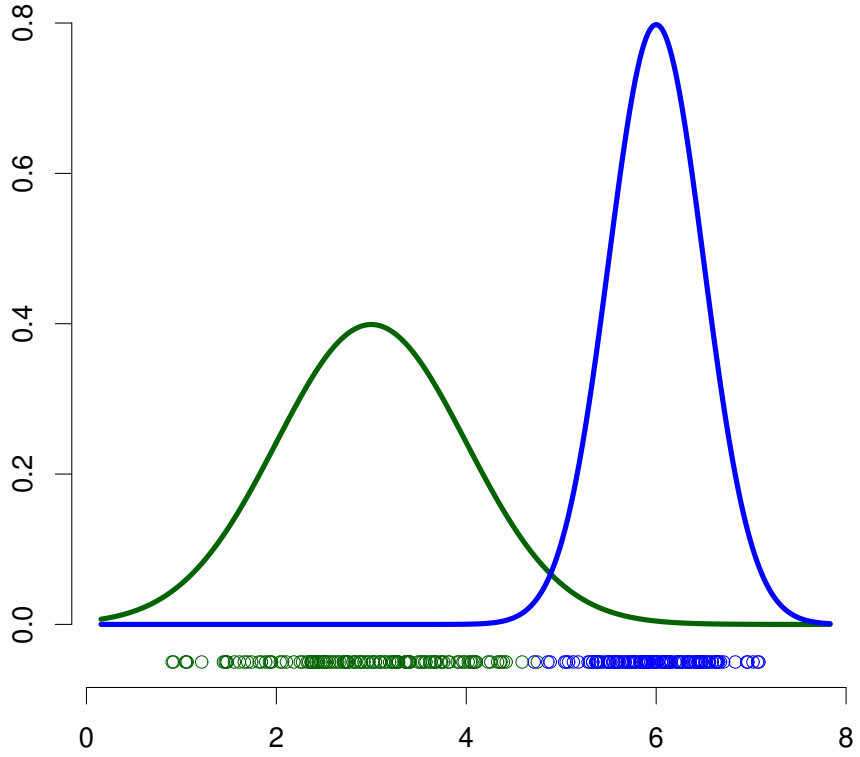
$\Rightarrow$ compute $\hat{y}_{ML} = \text{argmax}_{y \in \{1,\ldots,C\}}\, P(\boldsymbol{x}|\mathscr{h}_y, \Theta_y) = \text{argmax}_{y \in \{1,\ldots,C\}}\, \hat{f}_y(\boldsymbol{x})$

- $\hat{f}_y(\boldsymbol{x})$ is the estimated density for class y computed in $\boldsymbol{x}$

Maximum likelihood estimation: 1-dimensional case

Let x_y^l denote the l-th point in D_y

Parameter estimation:



$$\hat{\mu}_y = \frac{1}{|D_y|} \sum_{l=1}^{|D_y|} x_y^l$$

$$\hat{\sigma}_y^2 = \frac{1}{|D_y|} \sum_{l=1}^{|D_y|} (x_y^l - \hat{\mu}_y)^2$$

$$\hat{f}_y(x) = \frac{1}{\sqrt{2\pi\hat{\sigma}_y^2}} \exp\left[-\frac{1}{2}\left(\frac{x-\hat{\mu}_y}{\hat{\sigma}_y}\right)^2\right]$$

Building Gaussian MLE estimators (1-dim. case)

Assume m independent examples from a given class: $\{x^1, \ldots, x^m\}$

$$L = P(D|\hbar, \Theta) = \prod_{l=1}^{m} P(x^l|\hbar, \Theta)$$

Modeling assumption: the data is normally distributed $\Rightarrow \Theta = \{\mu, \sigma^2\}$

We can either maximize the likelihood L:

$$L = \prod_{l=1}^{m} \frac{1}{\sqrt{2\pi\sigma^2}} \exp^{\left[-\frac{1}{2}\left(\frac{x^l-\mu}{\sigma}\right)^2\right]}$$

or, equivalently, the log-likelihood $\ln L$:

$$\ln L = \sum_{l=1}^{m} \ln \frac{1}{\sqrt{2\pi}} + \ln \frac{1}{\sqrt{\sigma^2}} - \frac{1}{2\sigma^2}(x^l - \mu)^2$$

$$\frac{\partial \ln L}{\partial \mu} = \sum_{l=1}^{m} \frac{(x^l-\mu)}{\sigma^2}$$

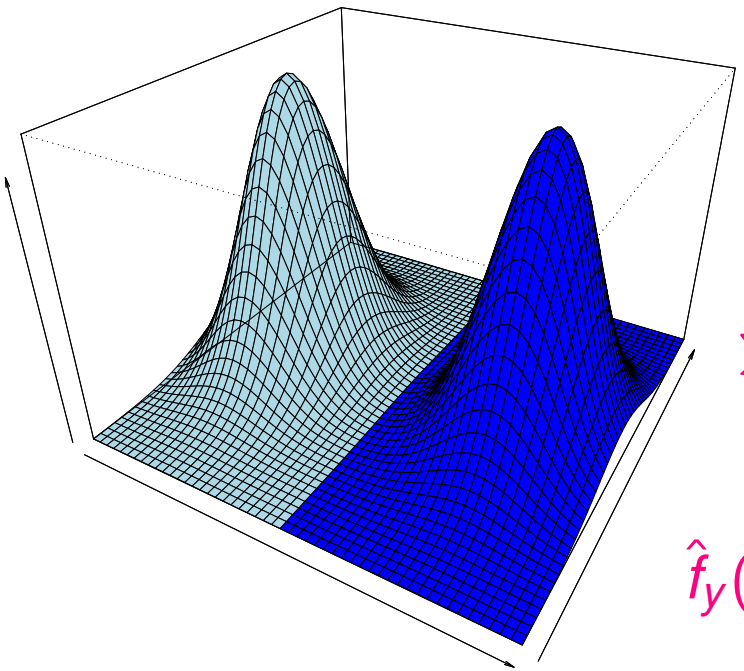
$$\left.\frac{\partial \ln L}{\partial \mu}\right|_{\mu=\hat{\mu}} = 0 \Rightarrow \hat{\mu} = \frac{1}{m}\sum_{l=1}^{m} x^l = \bar{X}$$

$$\left.\frac{\partial \ln L}{\partial \sigma^2}\right|_{\sigma^2=\hat{\sigma}^2} = 0 \Rightarrow \hat{\sigma}^2 = \frac{1}{m}\sum_{l=1}^{m}(x^l - \hat{\mu})^2$$

Maximum likelihood estimation: d-dimensional case

Let $\boldsymbol{x}_y^l$ denote the l-th vector in D_y

Parameter estimation:



$$\hat{\boldsymbol{\mu}}_y = \frac{1}{|D_y|}\sum_{l=1}^{|D_y|}\boldsymbol{x}_y^l$$

$$\hat{\boldsymbol{\Sigma}}_y = \frac{1}{|D_y|}\sum_{l=1}^{|D_y|}(\boldsymbol{x}_y^l - \hat{\boldsymbol{\mu}}_y)(\boldsymbol{x}_y^l - \hat{\boldsymbol{\mu}}_y)^T$$

$$\hat{f}_y(\boldsymbol{x}) = \frac{1}{\sqrt{(2\pi)^d|\hat{\boldsymbol{\Sigma}}_y|}}\exp\left[-\frac{1}{2}(\boldsymbol{x}-\hat{\boldsymbol{\mu}}_y)^T\hat{\boldsymbol{\Sigma}}_y^{-1}(\boldsymbol{x}-\hat{\boldsymbol{\mu}}_y)\right]$$

Contours associated to the covariance matrix

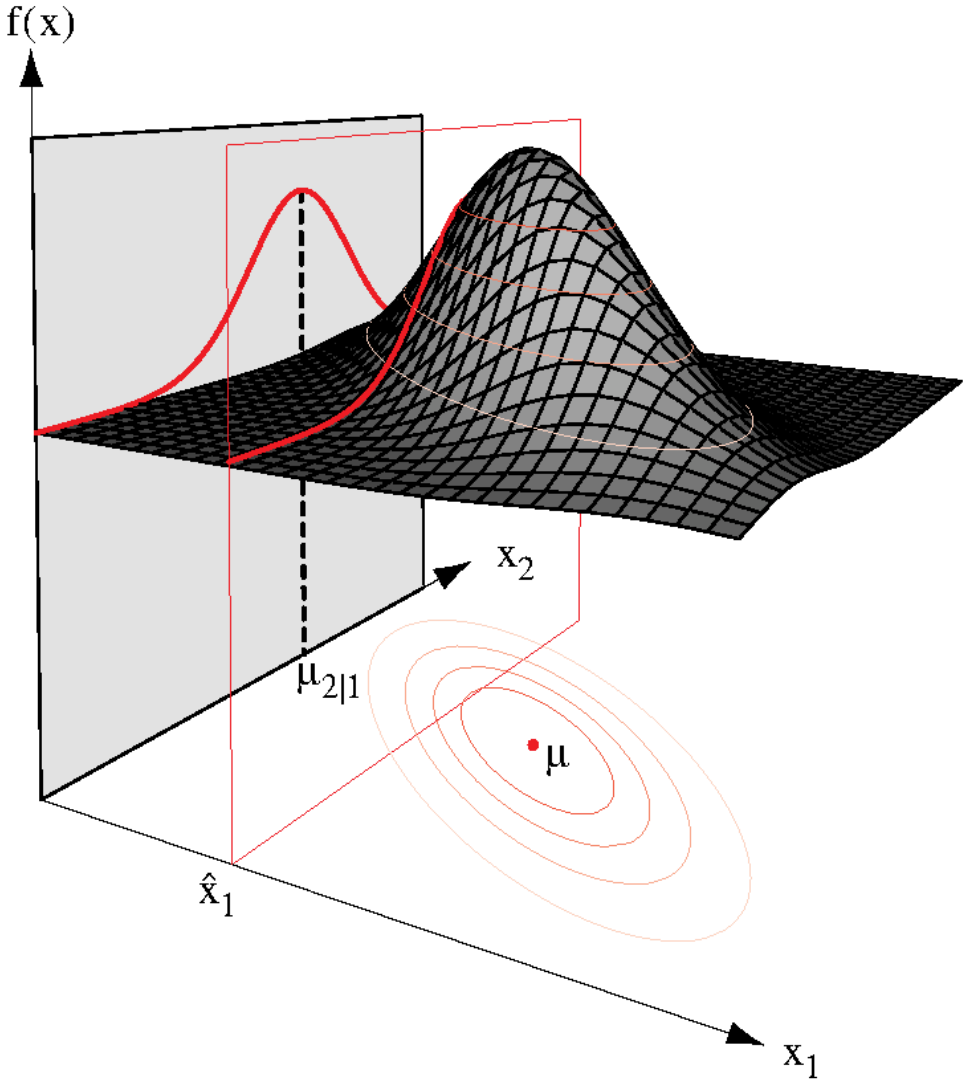
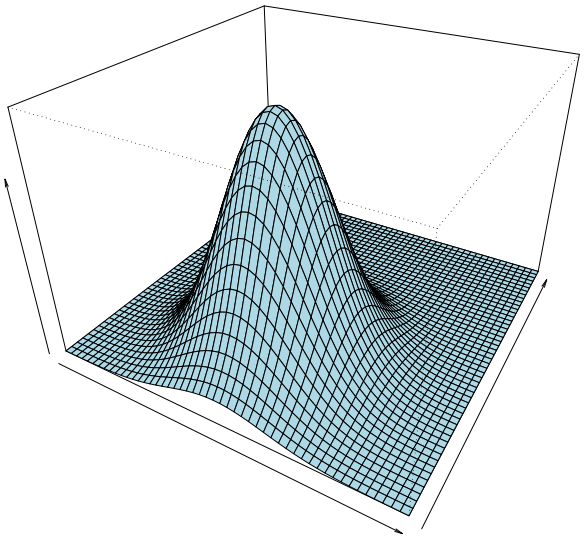
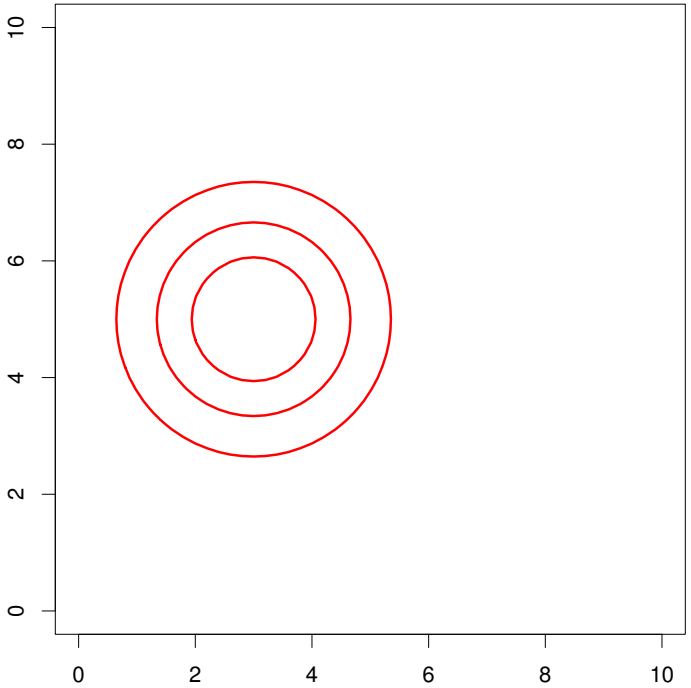


Illustration from *Pattern Classification* 2001, John Wiley and Sons, Inc.

Identity covariance matrix

Uncorrelated features, same variance in each dimension

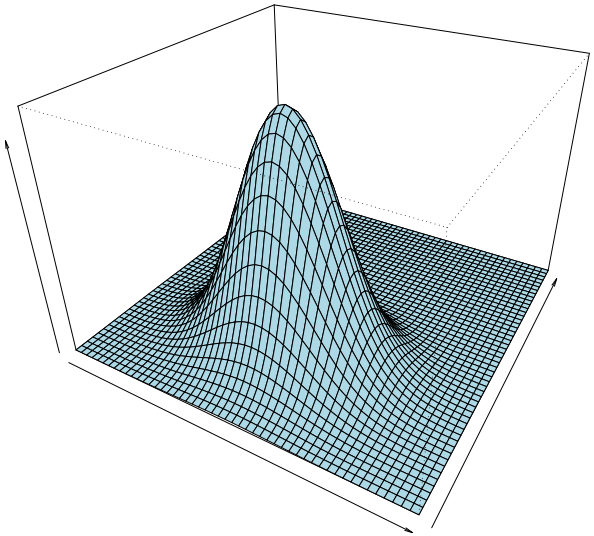


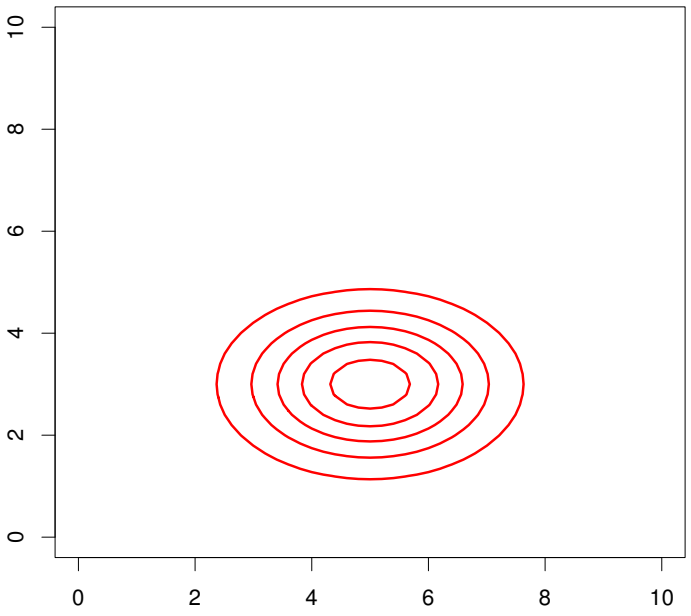


$\boldsymbol{\Sigma} = \sigma^2 \boldsymbol{I}$, for instance: $\boldsymbol{\Sigma} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Diagonal Covariance Matrix

Uncorrelated features

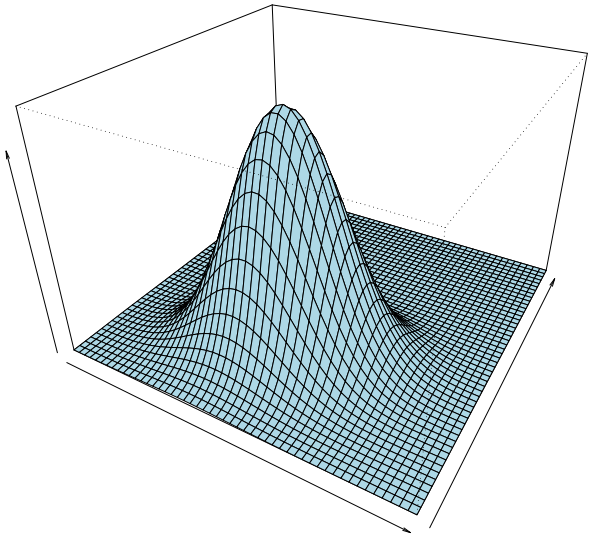


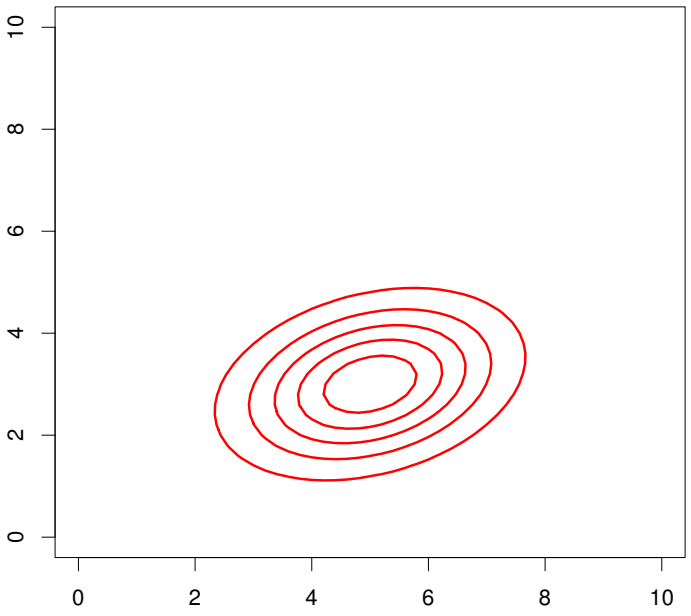


$\Sigma = diag\{\sigma_1^2, \ldots \sigma_d^2\}$ for instance: $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

Full Covariance Matrix

Slightly correlated features

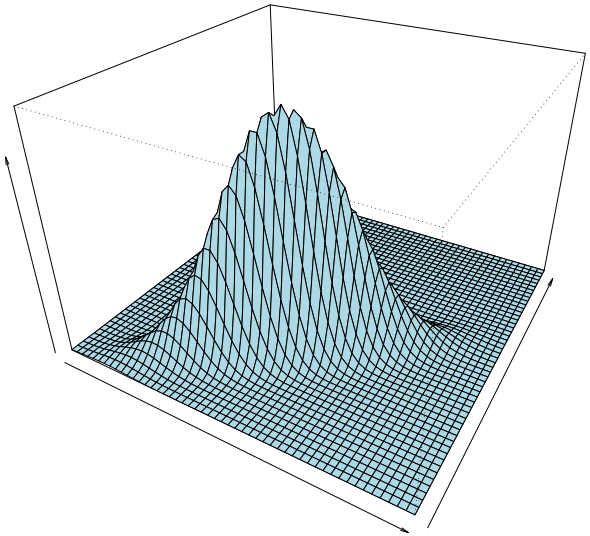


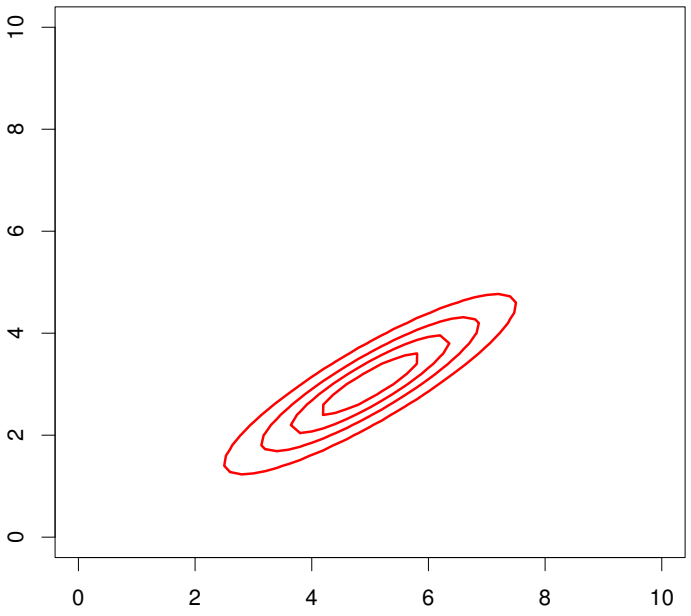


$$\Sigma = \begin{bmatrix} 1 & 0.4 \\ 0.4 & 2 \end{bmatrix}$$

Full Covariance Matrix

Highly correlated features

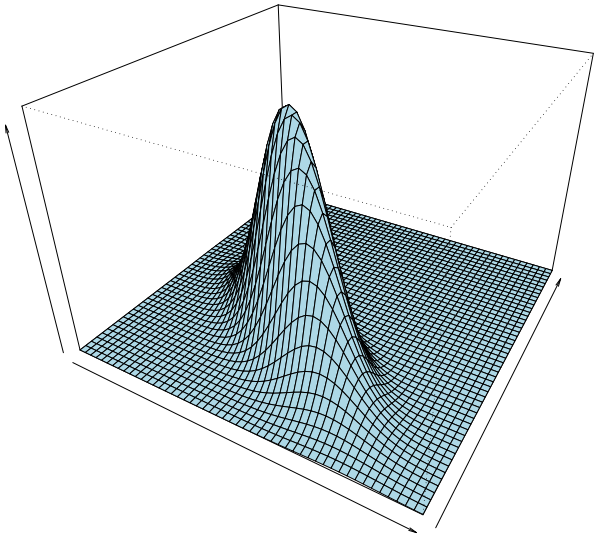


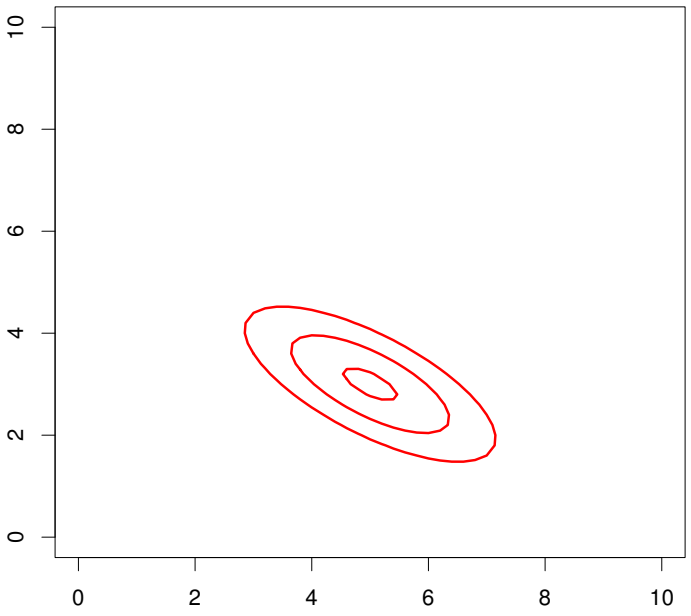


$$\Sigma = \begin{bmatrix} 1 & 1.25 \\ 1.25 & 2 \end{bmatrix}$$

Full Covariance Matrix

Negatively correlated features





$$\Sigma = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

Gaussian Naive Bayes Model

$$\boldsymbol{\mu}_y : \quad d \text{ parameters} \qquad \boldsymbol{\Sigma}_y : \quad \begin{cases} \text{Full:} & \mathcal{O}(d^2) \text{ parameters} \\ \text{Diagonal:} & d \text{ parameters} \\ \text{Identity:} & 1 \text{ parameter} \end{cases}$$

Diagonal assumption $\Rightarrow$ Naive Bayes assumption in $\mathbb{R}^d$

- uncorrelated features: independence assumption between coordinates
- reduced number of estimated parameters as compared to full covariance
- less accurate but more reliably estimated model with limited training data
- reduced computational cost (space and time):

$$\hat{f}_y(\boldsymbol{x}) = \frac{1}{\sqrt{(2\pi)^d |\hat{\boldsymbol{\Sigma}}_y|}} \exp\left[-\frac{1}{2}(\boldsymbol{x}-\hat{\boldsymbol{\mu}}_y)^T \hat{\boldsymbol{\Sigma}}_y{}^{-1}(\boldsymbol{x}-\hat{\boldsymbol{\mu}}_y)\right] = \prod_{j=1}^{d} \frac{1}{\sqrt{2\pi\hat{\sigma}_y[j]^2}} \exp\left[-\frac{1}{2}\left(\frac{x[j]-\hat{\mu}_y[j]}{\hat{\sigma}_i[j]}\right)^2\right]$$

Decision Boundary

Define a function $g(\boldsymbol{x})$ such that $g(\boldsymbol{x}) = 0$ whenever $\boldsymbol{x}$ lies on the decision boundary (2 classes)

MAP Classifier Decision Boundary

$\boldsymbol{x}$ on the decision boundary $\Leftrightarrow P(\boldsymbol{x}|y=1)P(y=1) = P(\boldsymbol{x}|y=2)P(y=2)$

We can define:

$$g(\boldsymbol{x}) = P(\boldsymbol{x}|y=1)P(y=1) - P(\boldsymbol{x}|y=2)P(y=2)$$

or:

$$g(\boldsymbol{x}) = g_1(\boldsymbol{x}) - g_2(\boldsymbol{x}) \text{ with } g_y(\boldsymbol{x}) = \ln P(\boldsymbol{x}|y) + \ln P(y)$$

Gaussian Decision Boundary

General case: arbitrary covariance matrices

$$
\begin{aligned}
P(\boldsymbol{x}|y) &\sim N(\boldsymbol{\mu}_y, \Sigma_y) \\
g_y(\boldsymbol{x}) &= \ln P(\boldsymbol{x}|y) + \ln P(y) \\
&= -\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_y)^T \Sigma_y^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_y) - \frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |\Sigma_y| + \ln P(y)
\end{aligned}
$$

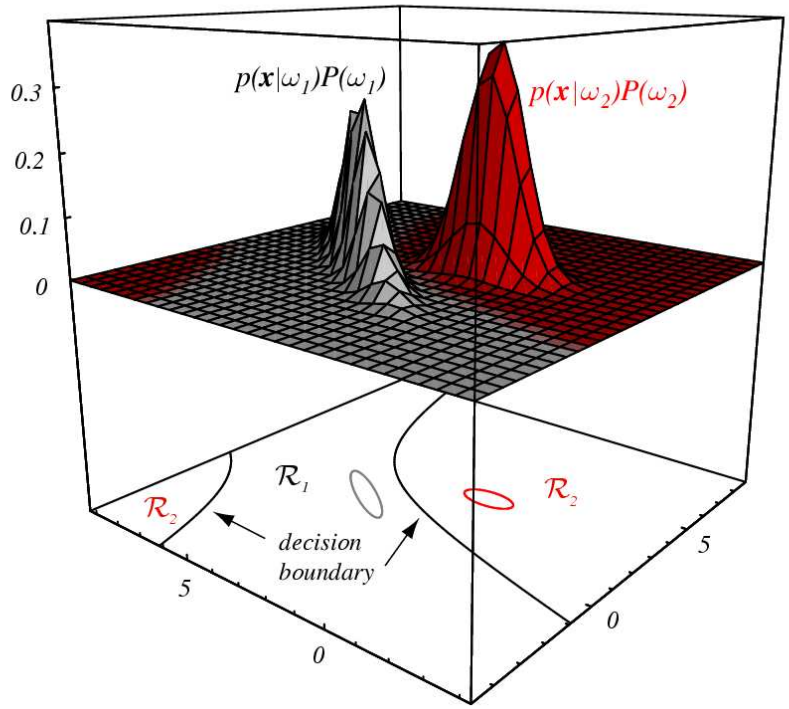
Note: $(\boldsymbol{x} - \boldsymbol{\mu}_y)^T \Sigma_y^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_y)$ is the Mahalanobis distance: distance to the mean weighted by the class variance. It follows:

$$
\begin{aligned}
g(\boldsymbol{x}) &= g_1(\boldsymbol{x}) - g_2(\boldsymbol{x}) \\
&= -\frac{1}{2} \boldsymbol{x}^T \left(\Sigma_1^{-1} - \Sigma_2^{-1} \right) \boldsymbol{x} && \text{Quadratic term} \\
&\quad + \boldsymbol{x}^T \left(\Sigma_1^{-1} \boldsymbol{\mu}_1 - \Sigma_2^{-1} \boldsymbol{\mu}_2 \right) && \text{Linear term} \\
&\quad + w_0 && \text{Constant term}
\end{aligned}
$$

with $w_0 = \frac{1}{2} \ln \frac{|\Sigma_2|}{|\Sigma_1|} - \frac{1}{2} \left(\boldsymbol{\mu}_1^T \Sigma_1^{-1} \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2^T \Sigma_2^{-1} \boldsymbol{\mu}_2 \right) + \ln \frac{P(y=1)}{P(y=2)}$

Gaussian Decision Boundary

General case: decision boundary are hyperquadrics



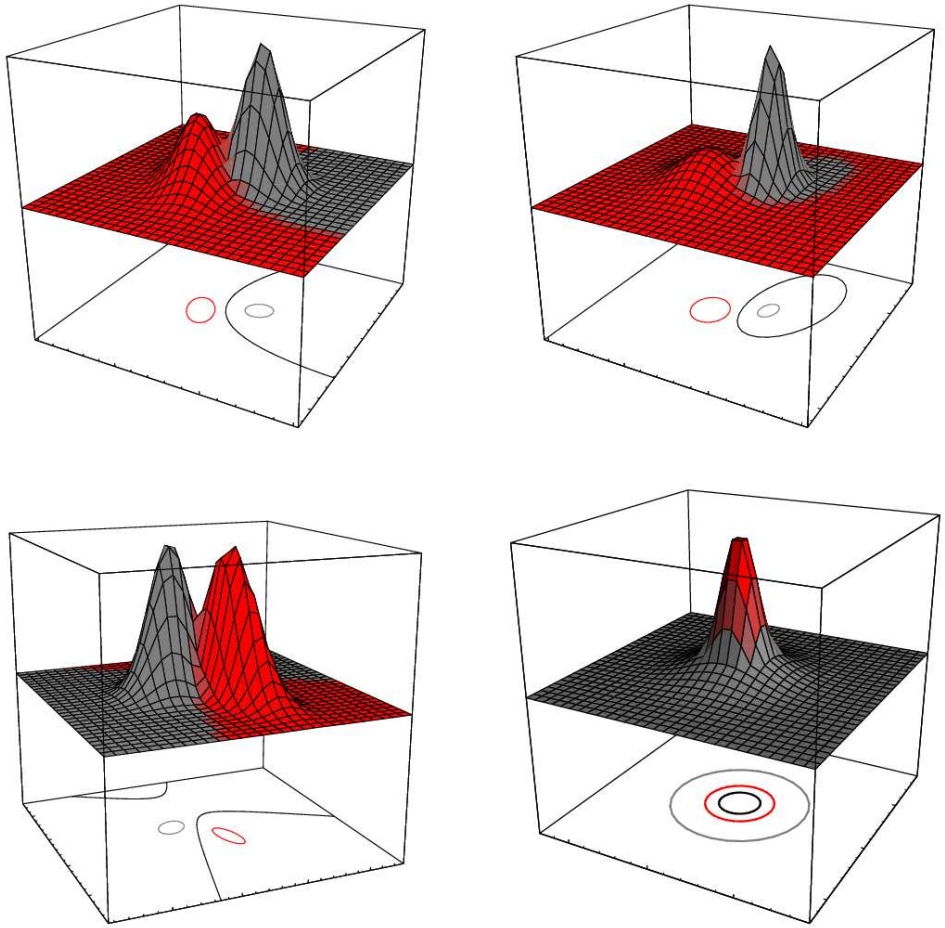


Illustration Note:

classes y are denoted ω_1 or ω_2

Illustrations from *Pattern Classification* 2001, John Wiley and Sons, Inc.

Gaussian Decision Boundary

Equal covariance matrices $\Sigma_1 = \Sigma_2 = \Sigma$

The quadratic term vanishes $-\frac{1}{2}\boldsymbol{x}^T\left(\Sigma_1^{-1} - \Sigma_2^{-1}\right)\boldsymbol{x} = 0$ and the decision boundary becomes linear

$$g(\boldsymbol{x}) = \boldsymbol{x}^T\left(\Sigma^{-1}(\mu_1 - \mu_2)\right) + w_0'$$

$$\text{with } w_0' = -\frac{1}{2}\left(\mu_1^T\Sigma^{-1}\mu_1 - \mu_2^T\Sigma^{-1}\mu_2\right) + \ln\frac{P(y=1)}{P(y=2)}$$

and can be rewritten as

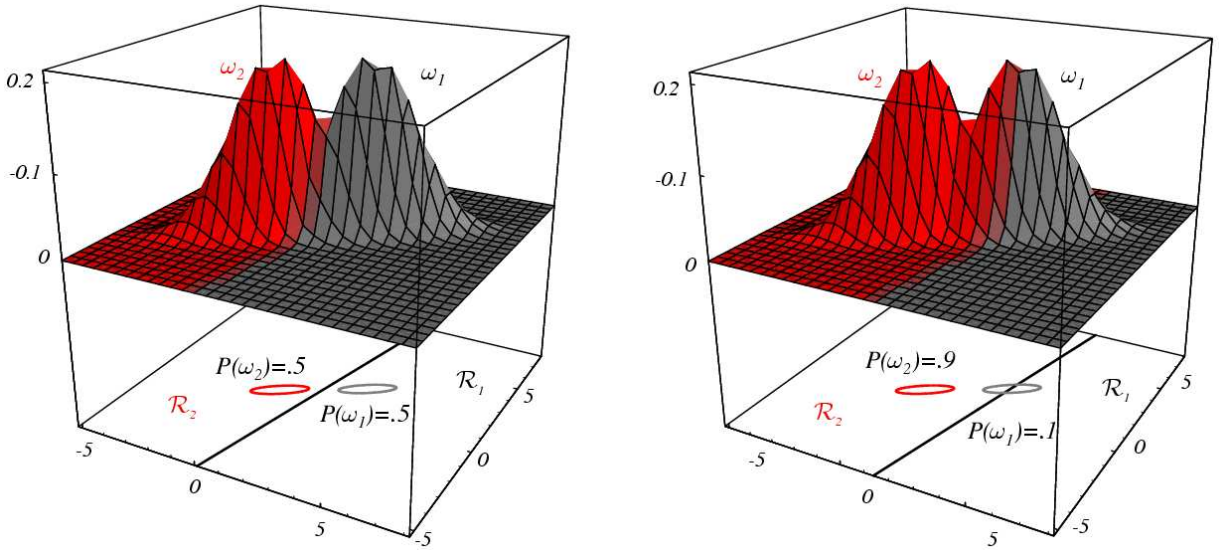
$$g(\boldsymbol{x}) = \boldsymbol{w}^T(\boldsymbol{x} - \boldsymbol{x}_0)$$

$$\text{with } \boldsymbol{w} = \Sigma^{-1}(\mu_1 - \mu_2)$$

$$\text{and } \boldsymbol{x}_0 = \frac{\mu_1 + \mu_2}{2} - \frac{\ln\frac{P(y=1)}{P(y=2)}}{(\mu_1 - \mu_2)^T\Sigma^{-1}(\mu_1 - \mu_2)}(\mu_1 - \mu_2)$$

Gaussian Decision Boundary

Equal covariance matrices: decision boundaries are hyperplanes



$\boldsymbol{w} = \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$ is generally not in the direction of $\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2 \Rightarrow$ the decision boundary is generally not orthogonal to the line connecting the class means
However, it intersects this line in x_0, which is halfway between the means when the class priors are equal

$$\boldsymbol{x}_0 = \frac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2}{2} - \frac{\ln \frac{P(y=1)}{P(y=2)}}{(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$$

Illustrations from *Pattern Classification* 2001, John Wiley and Sons, Inc.

Gaussian Decision Boundary

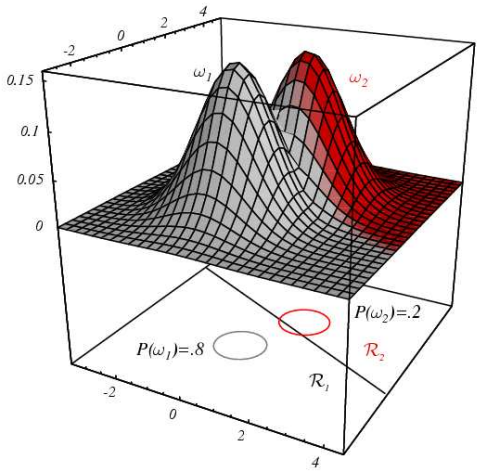
Uncorrelated features with equal variance in each dimension $\boldsymbol{\Sigma}_1 = \boldsymbol{\Sigma}_2 = \sigma^2 \boldsymbol{I}$

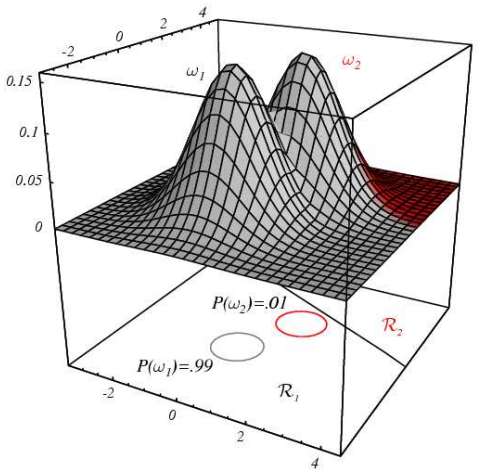
$$\boldsymbol{\Sigma}^{-1} = 1/\sigma^2 \boldsymbol{I} \Rightarrow g(\boldsymbol{x}) = \boldsymbol{w}^T(\boldsymbol{x} - \boldsymbol{x}_0)$$

with $\boldsymbol{w} = \boldsymbol{\mu}_1 - \boldsymbol{\mu}_2$ Note: $1/\sigma^2$can be seen as a scaling factor

and $\boldsymbol{x}_0 = \dfrac{\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2}{2} - \dfrac{\sigma^2}{\|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2\|^2} \ln \dfrac{P(y=1)}{P(y=2)} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)$

The decision boundary is a hyperplane orthogonal to the line connecting the means





Illustrations from *Pattern Classification* 2001, John Wiley and Sons, Inc.

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2 Fisher Linear Discriminant
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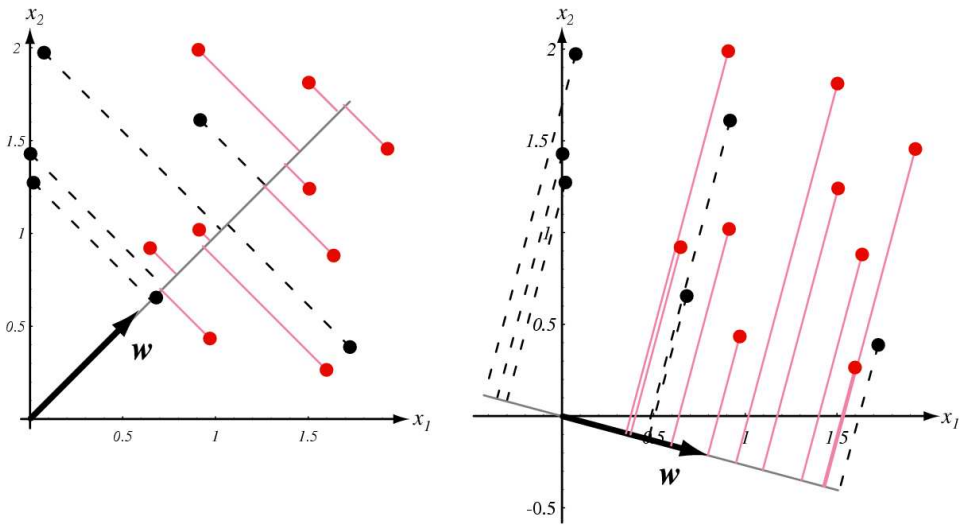
Optimal 1-dimensional projection

We consider the problem of projecting some data in $\mathbb{R}^d$ onto a line (1 dimensional space). Discriminant analysis looks for the **optimal projection** to discriminate between the classes

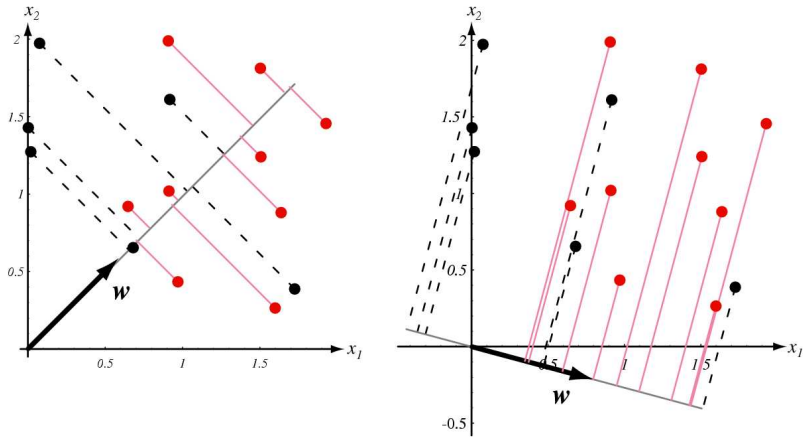


Finding the optimal projection has a strong relation with the classification problem

Illustrations from *Pattern Classification* 2001, John Wiley and Sons, Inc.

Fisher Linear Discriminant

- $D = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n\}$ be a training set of n points in $\mathbb{R}^d$ and D_y be the subset of the n_y points belonging to class y
- For any d-dimensional vector $\boldsymbol{w}$, the scalar $\boldsymbol{w}^T\boldsymbol{x}$ denotes the projection of $\boldsymbol{x}$ onto a line in the direction of $\boldsymbol{w}$



The sample mean of class y is $\boldsymbol{m}_y = \frac{1}{n_y}\sum_{\boldsymbol{x}\in D_y}\boldsymbol{x}$

The projected mean of class y is $\tilde{m}_y = \frac{1}{n_y}\sum_{\boldsymbol{x}\in D_y}\boldsymbol{w}^T\boldsymbol{x} = \boldsymbol{w}^T\boldsymbol{m}_y$

The scatter for projected points of class y is $\tilde{s}_y^2 = \sum_{\boldsymbol{x}\in D_y}(\boldsymbol{w}^T\boldsymbol{x} - \tilde{m}_y)^2$

We look for the vector $\boldsymbol{w}$ maximizing:

$$J(\boldsymbol{w}) = \frac{(\tilde{m}_1 - \tilde{m}_2)^2}{\tilde{s}_1^2 + \tilde{s}_2^2} = \frac{\text{Squared distance between projected means}}{\text{Variances of projected points}}$$

Within and between class scatter matrices

Scatter matrix: $\boldsymbol{S}_y = \sum_{\boldsymbol{x} \in D_y} (\boldsymbol{x} - \boldsymbol{m}_y)(\boldsymbol{x} - \boldsymbol{m}_y)^T$
Within class scatter matrix: $\boldsymbol{S}_W = \boldsymbol{S}_1 + \boldsymbol{S}_2$
$\Rightarrow$ proportional to the covariance of the *pooled* data in $\mathbb{R}^d$
Between class scatter matrix: $\boldsymbol{S}_B = (\boldsymbol{m}_1 - \boldsymbol{m}_2)(\boldsymbol{m}_1 - \boldsymbol{m}_2)^T$

Since

$$\tilde{s}_y^2 = \sum_{\boldsymbol{x} \in D_y} (\boldsymbol{w}^T \boldsymbol{x} - \boldsymbol{w}^T \boldsymbol{m}_y)^2 = \sum_{\boldsymbol{x} \in D_y} \boldsymbol{w}^T (\boldsymbol{x} - \boldsymbol{m}_y)(\boldsymbol{x} - \boldsymbol{m}_y)^T \boldsymbol{w} = \boldsymbol{w}^T \boldsymbol{S}_y \boldsymbol{w}$$

$$(\tilde{m}_1 - \tilde{m}_2)^2 = (\boldsymbol{w}^T \boldsymbol{m}_1 - \boldsymbol{w}^T \boldsymbol{m}_2)^2 = \boldsymbol{w}^T (\boldsymbol{m}_1 - \boldsymbol{m}_2)(\boldsymbol{m}_1 - \boldsymbol{m}_2)^T \boldsymbol{w} = \boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w}$$

we can rewrite

$$J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w}}{\boldsymbol{w}^T (\boldsymbol{S}_1 + \boldsymbol{S}_2) \boldsymbol{w}} = \frac{\boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w}}{\boldsymbol{w}^T \boldsymbol{S}_W \boldsymbol{w}}$$

Finding the optimal projection

A vector $\boldsymbol{w}$ maximizing $J(\boldsymbol{w}) = \frac{\boldsymbol{w}^T \boldsymbol{S}_B \boldsymbol{w}}{\boldsymbol{w}^T \boldsymbol{S}_W \boldsymbol{w}}$ must satisfy:

$$\boldsymbol{S}_B \boldsymbol{w} = \lambda \boldsymbol{S}_W \boldsymbol{w} \text{ for some constant } \lambda$$

This generalized eigenvalue problem can be rewritten, since $\boldsymbol{S}_W$ is generally non singular (when $n > d$), as a standard eigenvalue problem:

$$\boldsymbol{S}_W^{-1} \boldsymbol{S}_B \boldsymbol{w} = \lambda \boldsymbol{w}$$

For any $\boldsymbol{w}$, $\boldsymbol{S}_B \boldsymbol{w} = (\boldsymbol{m}_1 - \boldsymbol{m}_2)(\boldsymbol{m}_1 - \boldsymbol{m}_2)^T \boldsymbol{w} = (\boldsymbol{m}_1 - \boldsymbol{m}_2)\kappa$ with κ a scalar $\Rightarrow$ $\boldsymbol{S}_B \boldsymbol{w}$ is in the direction of $(\boldsymbol{m}_1 - \boldsymbol{m}_2)$

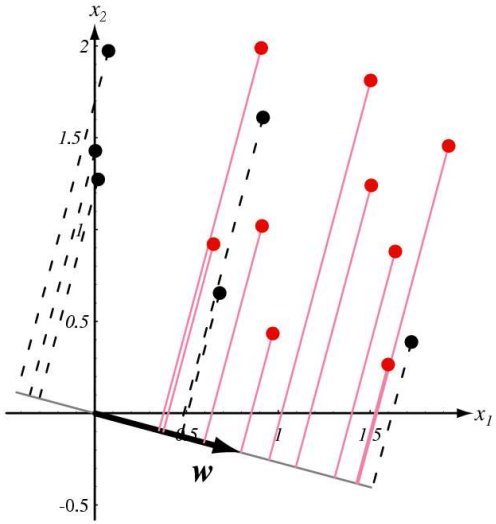
$$\boldsymbol{S}_W^{-1}(\boldsymbol{m}_1 - \boldsymbol{m}_2)\kappa = \lambda \boldsymbol{w} \Leftrightarrow \boldsymbol{w} \propto \boldsymbol{S}_W^{-1}(\boldsymbol{m}_1 - \boldsymbol{m}_2)$$

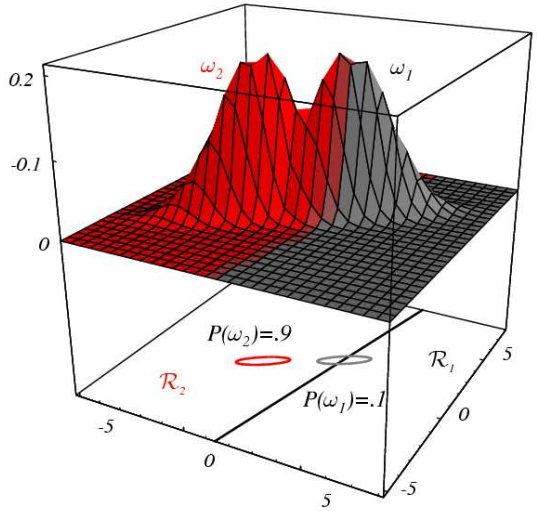
Only the direction of $\boldsymbol{w}$ matters, not its scale $\Rightarrow$ the solution is simply

$$\boxed{\boldsymbol{w} = \boldsymbol{S}_W^{-1}(\boldsymbol{m}_1 - \boldsymbol{m}_2)}$$

Fisher Discriminant Analysis as a Classifier

Once the data is projected on a line, a classifier is obtained by fixing a decision threshold w_0 on this line $\Rightarrow$ decision boundary: $\boldsymbol{w}^T\boldsymbol{x} + w_0 = 0$





$\boldsymbol{w} = \boldsymbol{S}_W^{-1}(\boldsymbol{m}_1 - \boldsymbol{m}_2)$ $\qquad$ $\boldsymbol{w} = \Sigma^{-1}(\mu_1 - \mu_2)$

- **F**isher **D**iscriminant **A**nalysis $\equiv$ **L**inear **D**iscriminant **A**nalysis $\equiv$ Gaussian classifier **whenever** $\hat{\Sigma} = \boldsymbol{S}_W$ and $\hat{\mu}_y = \boldsymbol{m}_y$
- The Gaussian classifier with arbitrary covariance matrices is also known as **Q**uadratic **D**iscriminant **A**nalysis

Illustrations from *Pattern Classification* 2001, John Wiley and Sons, Inc.

Summary

- **Gaussian classifiers**
  - Gaussian densities are powerful parametric techniques to model data in $\mathbb{R}^d$
  - Diagonal covariance assumption offers a trade-off between model complexity and accuracy
  - Gaussian classifiers define quadratic discriminants in general, and linear ones when class covariances are (assumed) equal
- **Fisher Discriminant Analysis**
  - FDA projects the data on a line to best discriminate between classes
  - FDA defines a linear classifier equivalent to a special case of Gaussian classifier